

MEDICINE HAT, AB, Canada

WMO: 712470

Lat: 50.0250N

Lon: 110.7172W

Elev: 2352

StdP: 13.49

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.6	-14.8	-30.0	1.1	-21.3	-22.3	1.8	-14.4	29.3	40.6	26.3	37.2	4.8	220	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	91.4	63.2	88.0	62.4	84.5	61.5	66.1	83.7	64.5	82.3	63.0	80.4	9.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	85.1	71.5	58.3	79.2	70.2	56.4	74.0	68.8	32.2	83.8	30.8	82.1	29.7	80.8	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.1	21.8	18.9	-28.6	97.5	5.4	2.5	-32.5	99.3	-35.6	100.7	-38.6	102.2	-42.5	104.0		
			-28.9	69.9	5.3	1.8	-32.7	71.2	-35.7	72.3	-38.7	73.3	-42.5	74.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.8	16.8	19.9	31.1	43.6	53.6	61.4	68.6	67.0	57.2	44.0	30.1	19.1	(6)
(7)		DBStd	21.43	16.52	15.24	14.30	9.08	8.20	6.24	5.88	6.55	8.52	9.34	13.19	15.11	(7)
(8)		HDD50	4557	1029	842	588	228	55	2	0	0	28	227	601	957	(8)
(9)		HDD65	8416	1494	1262	1050	644	362	142	29	56	259	650	1046	1422	(9)
(10)		CDD50	1943	0	0	3	34	167	344	578	526	244	42	5	0	(10)
(11)		CDD65	328	0	0	0	0	9	34	142	117	25	1	0	0	(11)
(12)		CDH74	4946	0	0	1	20	217	525	1963	1698	503	19	0	0	(12)
(13)		CDH80	1877	0	0	0	2	47	143	804	705	175	1	0	0	(13)
(14)	Wind	WSAvg	8.5	8.6	8.2	8.9	9.4	9.0	8.6	7.8	7.4	7.8	8.8	9.0	8.8	(14)
(15)	Precipitation	PrecAvg	13.7	0.8	0.5	0.7	1.1	1.7	2.2	1.7	1.3	1.5	0.7	0.6	0.7	(15)
(16)		PrecMax	21.8	2.4	1.7	1.7	3.8	3.6	5.2	3.5	3.7	8.0	1.5	1.6	2.1	(16)
(17)		PrecMin	8.7	0.1	0.0	0.0	0.0	0.3	0.4	0.3	0.2	0.1	0.0	0.1	0.2	(17)
(18)		PrecStd	3.2	0.6	0.4	0.4	0.8	0.9	1.4	0.9	0.9	1.5	0.4	0.4	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.3	54.6	66.1	76.7	85.5	89.9	95.7	95.6	91.0	76.7	64.2	51.9	(19)
(20)		MCWB	41.3	42.6	48.2	53.0	59.1	63.4	64.9	63.7	61.0	55.1	48.2	41.2	(20)	
(21)		2%	DB	45.0	47.3	60.0	70.6	80.0	84.0	91.8	91.3	85.1	69.9	57.3	46.8	(21)
(22)		MCWB	37.9	38.2	45.5	50.9	57.1	61.8	64.5	62.3	59.3	52.1	44.9	38.4	(22)	
(23)		5%	DB	41.5	42.8	54.8	65.8	75.2	79.9	88.1	87.8	80.0	64.6	52.2	42.7	(23)
(24)		MCWB	36.0	35.7	42.6	48.3	55.0	60.0	63.6	61.8	57.8	49.9	42.2	36.1	(24)	
(25)		10%	DB	38.7	39.1	49.8	60.6	70.6	76.1	84.6	83.7	74.6	59.7	47.4	39.2	(25)
(26)		MCWB	34.3	34.0	40.0	45.5	52.7	58.4	62.7	61.0	55.5	47.3	39.1	33.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.8	43.3	49.5	54.6	61.0	66.9	69.6	66.8	63.3	56.7	49.8	42.6	(27)
(28)		MDCB	49.1	52.8	64.7	74.0	80.5	82.8	87.7	85.4	83.9	73.5	61.7	50.0	(28)	
(29)		2%	WB	38.5	38.8	46.3	51.7	58.3	63.9	67.2	64.8	60.6	53.2	45.8	38.9	(29)
(30)		MDCB	44.2	46.4	58.5	69.2	75.9	79.0	84.5	83.9	80.5	67.3	56.5	45.7	(30)	
(31)		5%	WB	36.3	36.0	43.2	49.0	56.2	61.7	65.4	63.4	58.4	50.4	42.6	36.4	(31)
(32)		MDCB	41.0	41.9	53.7	63.9	72.3	76.2	82.7	82.4	77.4	62.9	51.2	42.5	(32)	
(33)		10%	WB	34.3	33.9	40.2	46.3	54.1	59.6	63.8	62.0	56.4	48.0	39.5	33.7	(33)
(34)		MDCB	38.8	39.1	49.1	59.1	68.2	73.3	81.1	80.2	73.2	58.8	46.9	39.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	18.6	19.2	20.7	23.3	23.6	22.6	26.4	26.8	25.4	22.9	19.4	18.3	(35)	
(36)		5% DB	MCDBR	19.0	22.8	28.4	32.4	31.6	29.3	31.9	33.9	35.2	30.5	25.5	20.0	(36)
(37)		MCWBR	14.0	15.8	16.5	16.7	14.0	12.3	12.0	12.6	15.1	16.1	15.4	13.9	(37)	
(38)		5% WB	MCDWR	18.9	21.8	27.2	30.0	28.0	26.3	28.1	29.2	32.6	29.2	24.3	19.9	(38)
(39)		MCWBR	14.4	15.8	16.4	16.0	13.1	12.2	11.8	11.6	14.5	15.9	15.3	14.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.271	0.267	0.301	0.359	0.383	0.373	0.372	0.394	0.344	0.303	0.270	0.257	(40)	
(41)		taud	2.389	2.469	2.439	2.275	2.252	2.344	2.361	2.267	2.388	2.485	2.479	2.408	(41)	
(42)		Ebn at Noon	249	284	290	280	277	281	279	266	270	265	250	239	(42)	
(43)		Edh at Noon	20	23	29	38	41	38	37	38	30	23	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	391	705	1131	1567	1858	2022	2199	1772	1266	784	437	305	(44)	
(45)		RadStd	37	60	79	109	130	119	109	112	104	80	32	27	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page